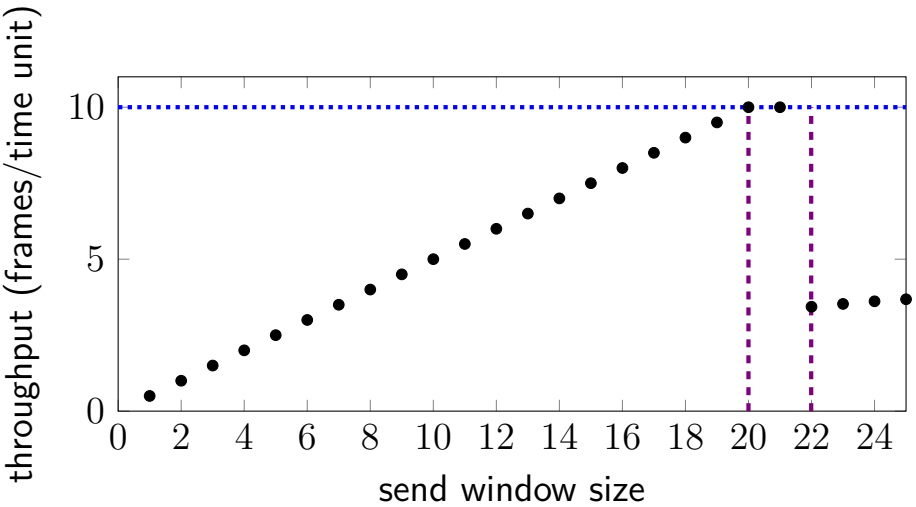
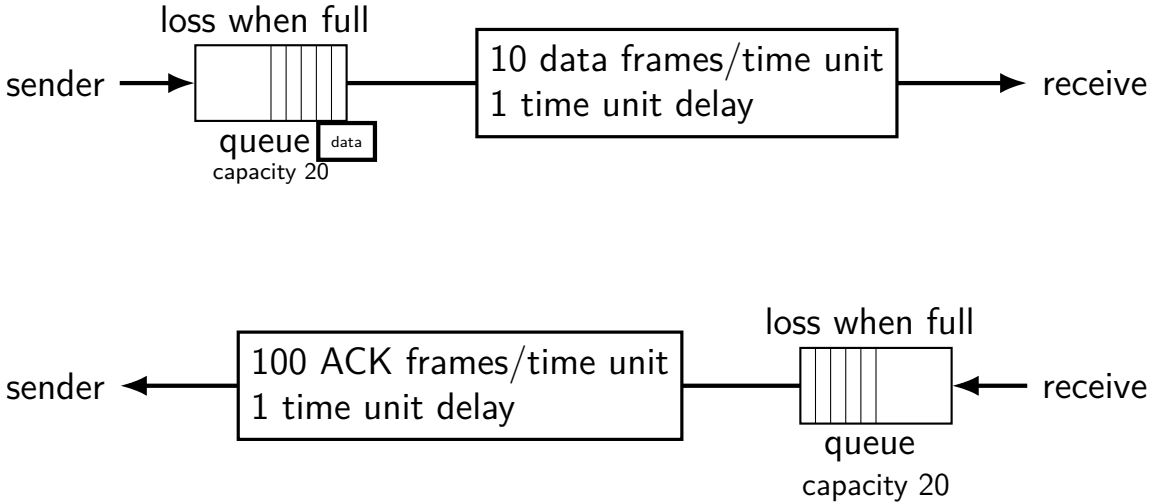
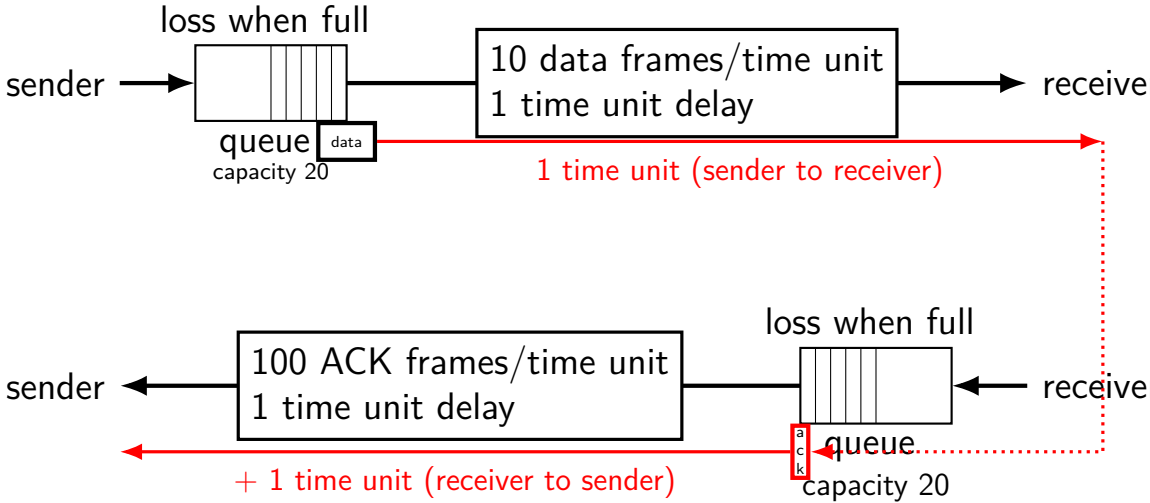
throughput and window size



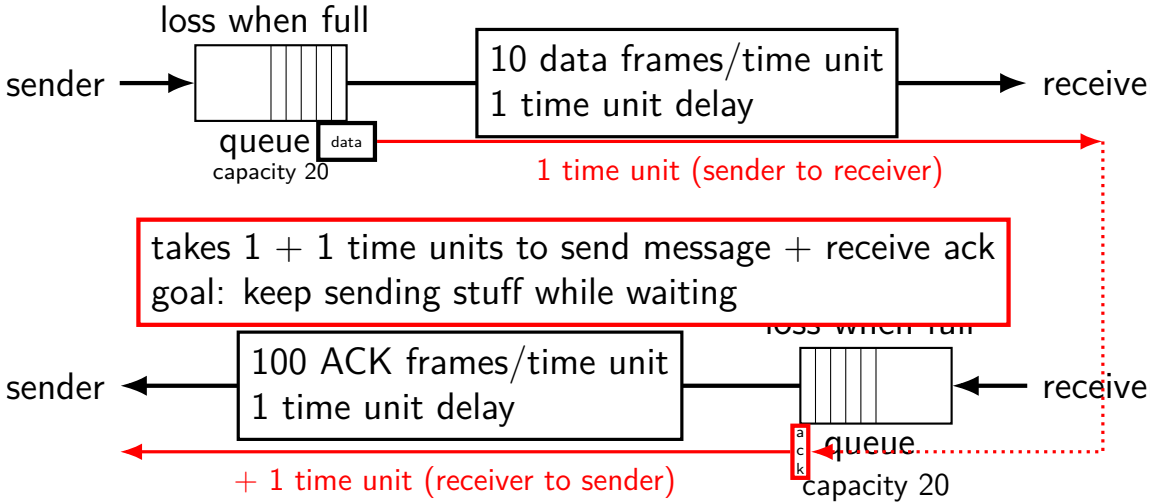
packet transit time



packet transit time



packet transit time



filling the pipe

round-trip time of 2 time units

from send data to receive ACK (assuming no queuing delay)

can send 10 data frames per time unit

= can send 20 data frames while waiting for ACK

filling the pipe

round-trip time of 2 time units

- from send data to receive ACK (assuming no queuing delay)

can send 10 data frames per time unit

= can send 20 data frames while waiting for ACK

"bandwidth-delay product"

- 10/time unit (banwidth) times 2 time unit (RTT = delay)

why optimal

in normal operation with window size W

- receive ACK for x (now $W - 1$ in flight)
- send packet $x + W$
- receive ACK for $x + 1$
- send packet $x + W + 1$
- ...

window size keeps W packets in flight

if links + queues can hold W packets — perfect!

number in flight on losses

window size W

let's say we lose packet x [only], sender might

- receive ACK for $x - 1$
- send packet $x + W - 1$
- receive ACK for x, x, x, ...
- resend packet x (guess it is lost)
- receive ACK for x, x, x, ...
- receive ACK for packet $x + W - 1$
- send packets $x + W$ through $x + W + W - 1$

number in flight on losses

window size W

let's say we lose packet x [only], sender might

- receive ACK for $x - 1$
- send packet $x + W - 1$
- receive ACK for x, x, x, ...
- resend packet x (guess it is lost)
- receive ACK for x, x, x, ...
- receive ACK for packet $x + W - 1$
- send packets $x + W$ through $x + W + W - 1$

lots of time where we are not sending packets
means network is underutilized

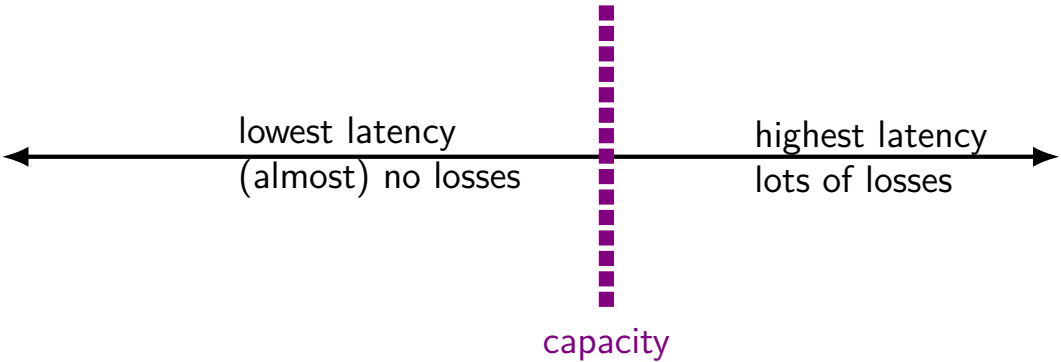
window size tweaking

window size imperfect proxy for # packets in flight

we'll ignore the difference for now

our goal for now: window size = number of packets to have in flight

finding window size empirically (1)



key insight

latency/loss rate increases when window size too big

latency/loss rate stable when window size not too big

for now, we'll focus on loss rate

- but you can do something similar with latency

try a bunch of things

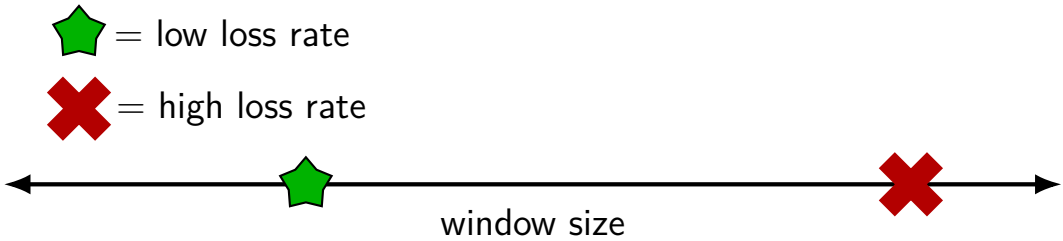
 = low loss rate

 = high loss rate

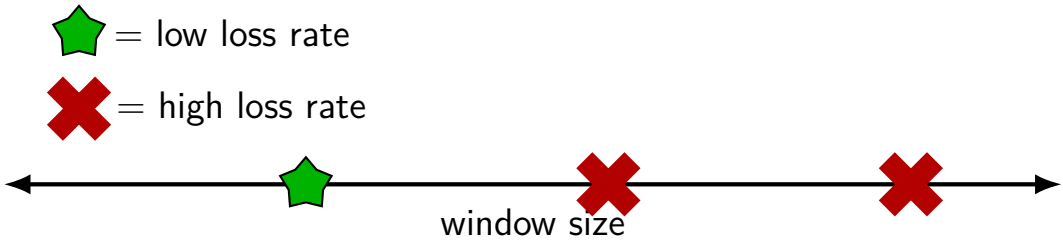


window size

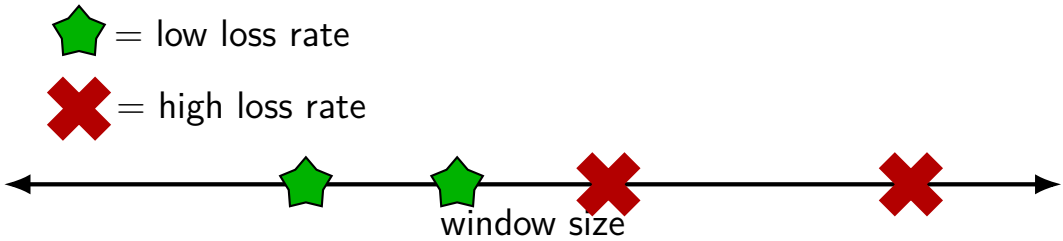
try a bunch of things



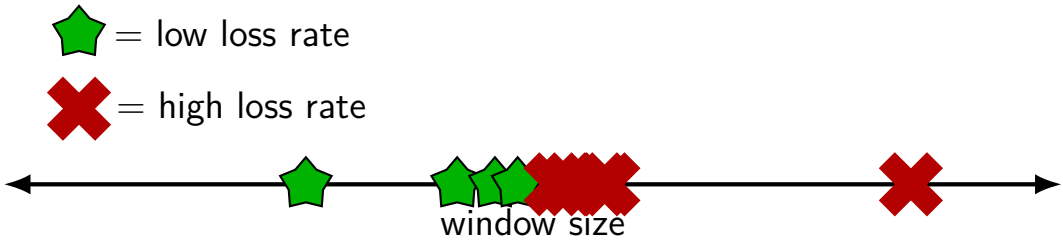
try a bunch of things



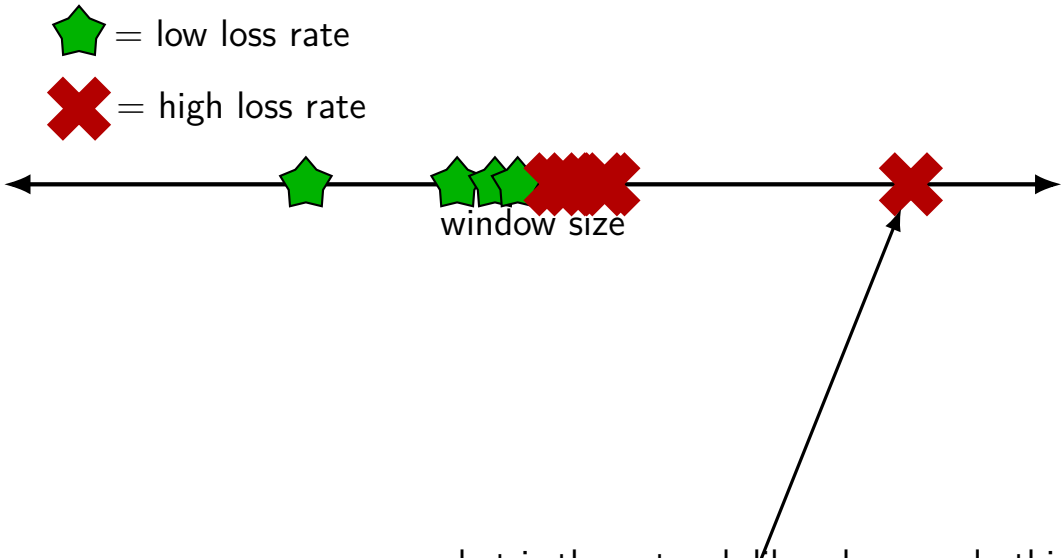
try a bunch of things



try a bunch of things



try a bunch of things



revisiting congestion collapse

Congestion Avoidance and Control

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In October of '86, the Internet had the first of what became a series of 'congestion collapses'. During this period, the data throughput from LBL to UC Berkeley (sites separated by 400 yards and three IMP hops) dropped from 32 Kbps to 40 bps. Mike Karels[1] and I were fascinated by this sudden factor-of-thousand drop in bandwidth and embarked on an investigation of why things had gotten so bad. We wondered, in particular,

fixes from Jacobson's 1987 paper

(*i*) round-trip-time variance estimation

(*ii*) exponential retransmit timer backoff

(*iii*) slow-start

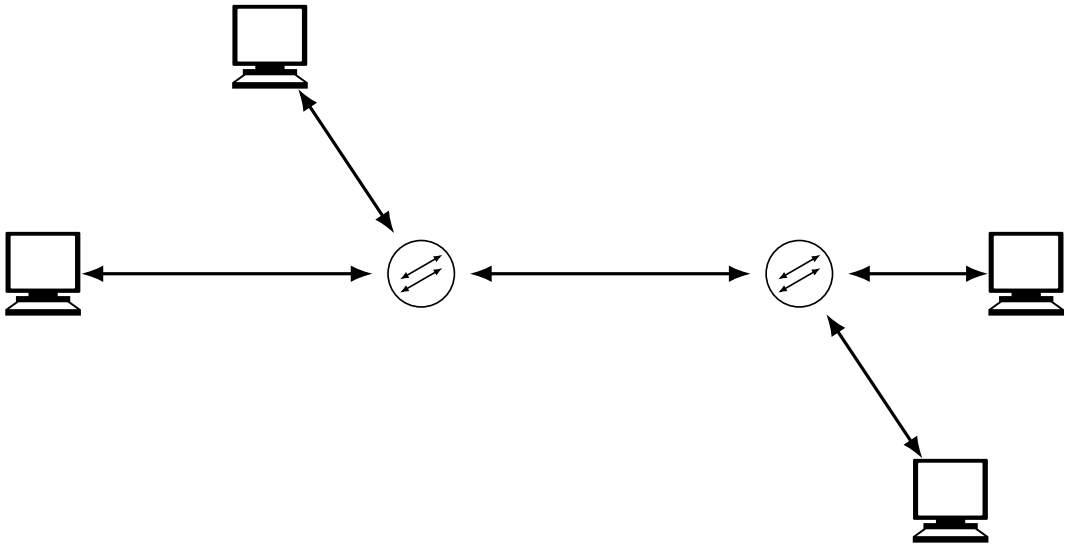
(*iv*) more aggressive receiver ack policy

(*v*) dynamic window sizing on congestion

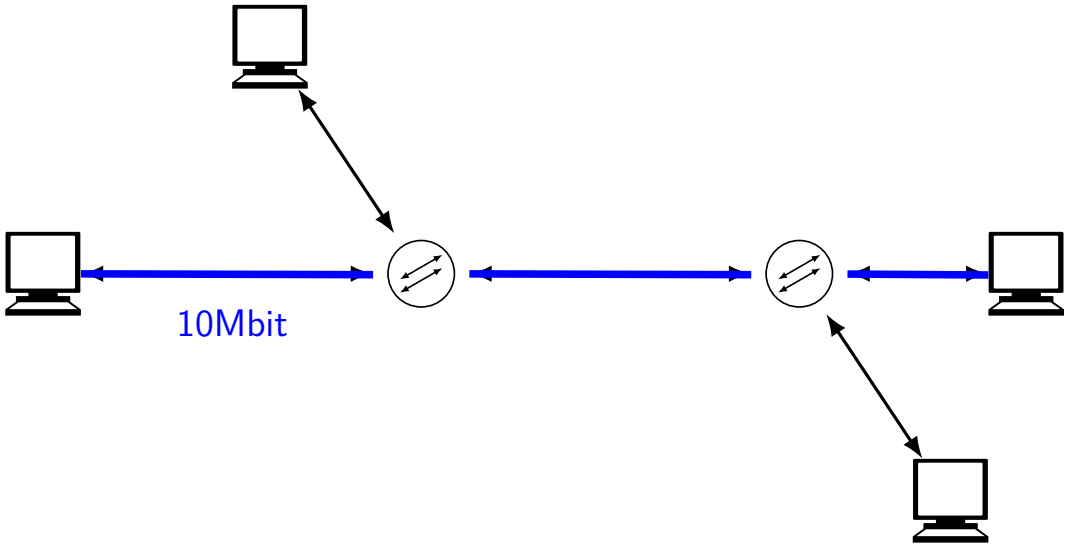
(*vi*) Karn's clamped retransmit backoff

(*vii*) fast retransmit

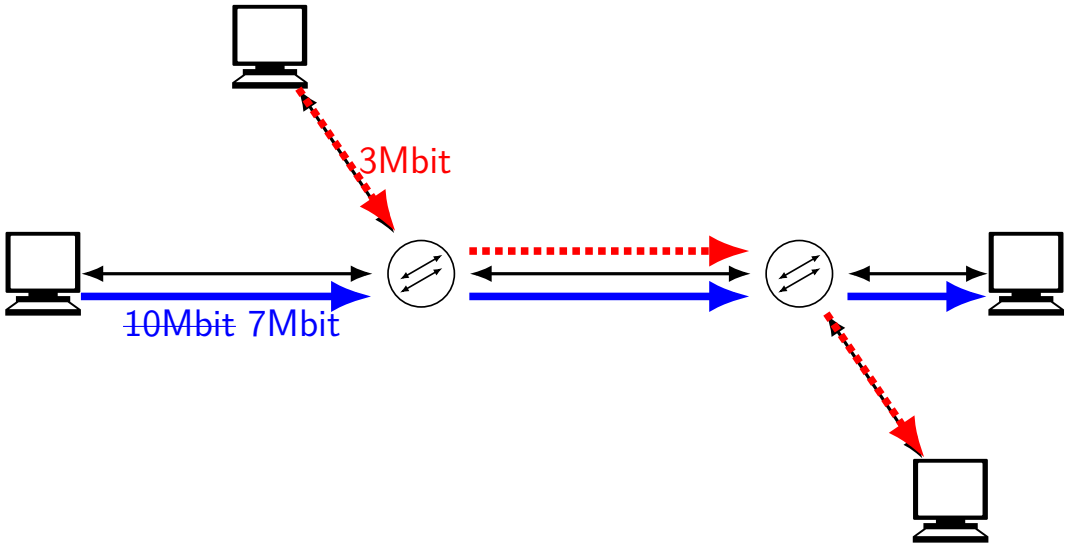
changing cross-traffic



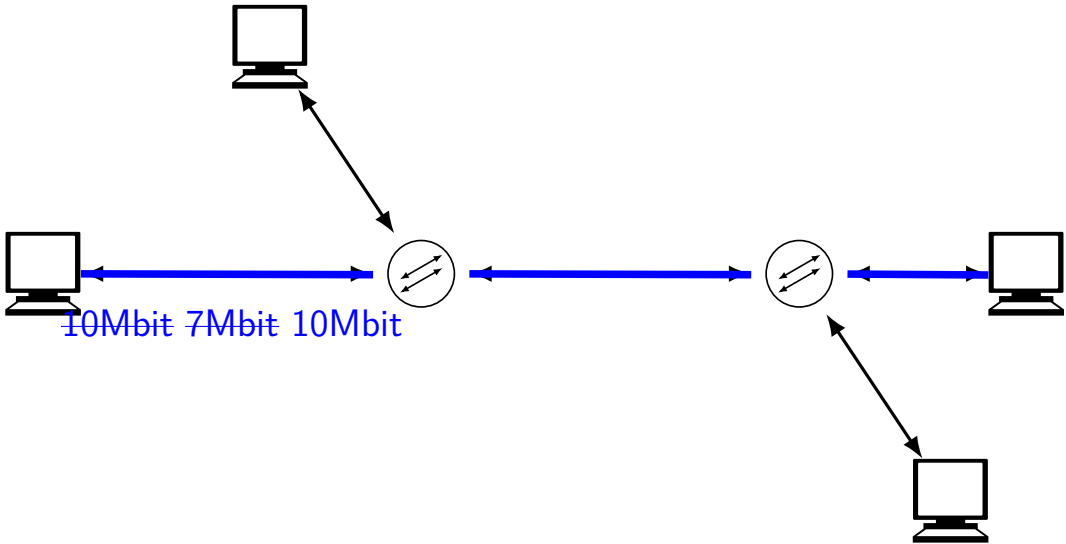
changing cross-traffic



changing cross-traffic



changing cross-traffic



adapting to cross-traffic

available bandwidth will change

previous example: 3Mbit lost/added from other flow

need to adapt to lost bandwidth

need to detect new available bandwidth

other flow's bandwidth?

for now, we'll pretend other flows don't react to us

later topic: what happens when both reacting?

fixes from Jacobson's 1987 paper

(*i*) round-trip-time variance estimation

(*ii*) exponential retransmit timer backoff

(*iii*) slow-start

(*iv*) more aggressive receiver ack policy

(*v*) dynamic window sizing on congestion

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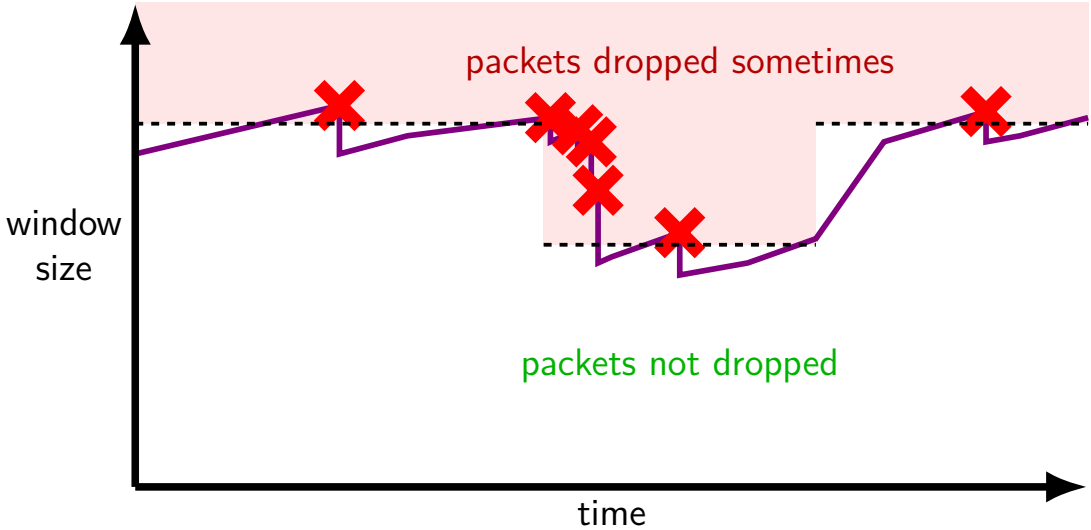
handling steady state

most of the time we should be at approx. correct window size

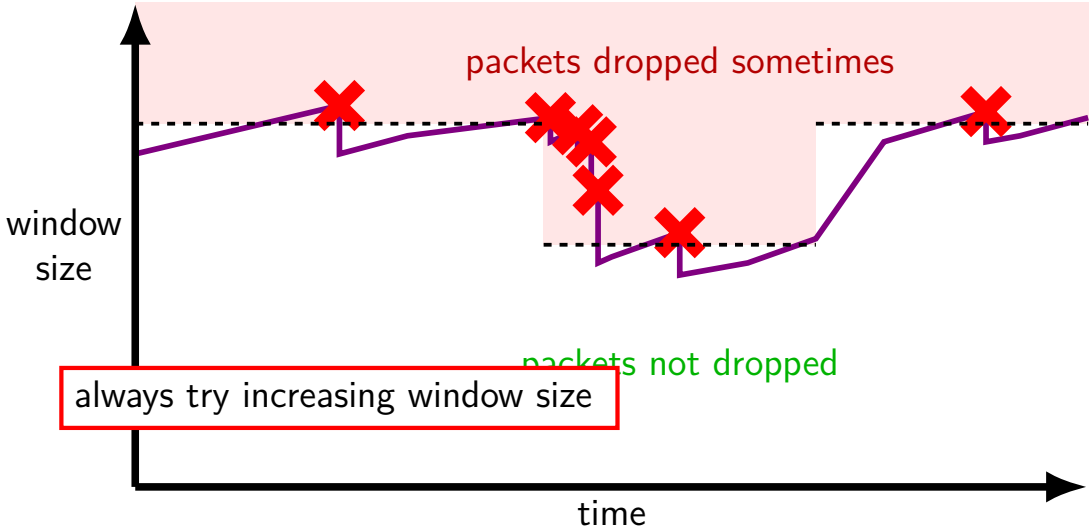
want to focus on how we react to changes

still going to use “experimentation” idea

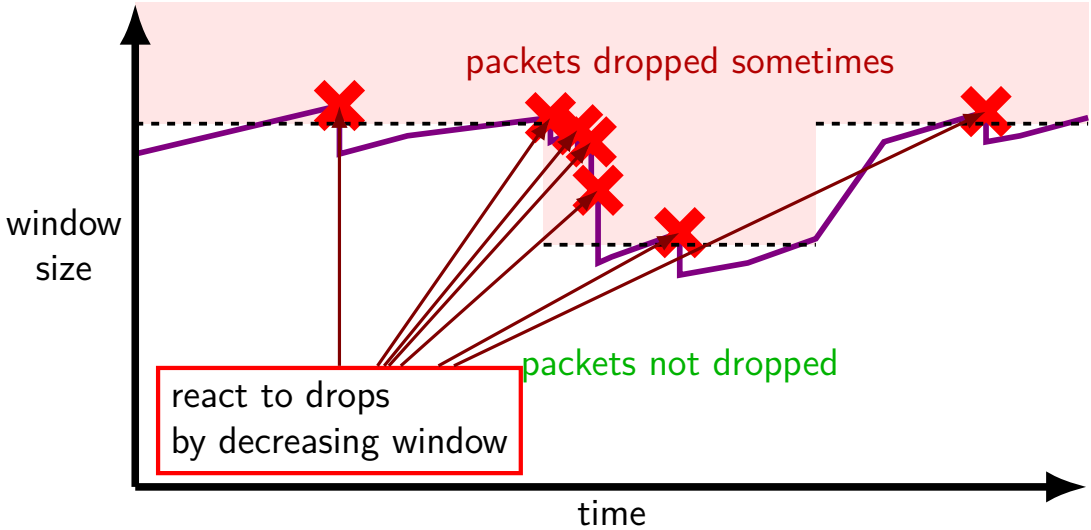
window size experimenting



window size experimenting



window size experimenting



increase/decrease strategy

default to increasing window size

react to packet drops by decreasing window size

- assumption: few “non-congestion” packet losses

increase/decrease strategy

default to increasing window size

react to packet drops by decreasing window size

assumption: few "non-congestion" packet losses

increase/decrease strategy

default to increasing window size

react to packet drops by decreasing window size

- assumption: few "non-congestion" packet losses

big topic: how fast to do each?

questions to help decide that:

- what happens if we increase too fast? too slow?
- what happens if we decrease too fast? too slow?

the overloaded switch

let's say switch can handle 50 packets/second

but has:

- 100 packets/second from test flow sending as fast as it can
- 10 packets/second from other session

expected *loss rate* (% packets lost)?

expected % test flow packets lost?

expected other session packets lost?

modeling who gets dropped

it kinda does matter...

sending in big bursts or spread out ("pacing")?

- bursts can overload queues even though average rate is low

how switch's queue works?

- queue size (handling bursts), way to choose what to drop

random or fixed intervals between sending?

modeling who gets dropped

it kinda does matter...

sending in big bursts or spread out ("pacing")?

- bursts can overload queues even though average rate is low

how switch's queue works?

- queue size (handling bursts), way to choose what to drop

random or fixed intervals between sending?

but we'll simplify, assuming—

- a flow's arrivals are randomly spaced
- drops hit packets at random
- queue is "pretty big"

the overloaded switch

let's say switch can handle 50 packets/second

but has:

- 100 packets/second from test flow (checking window size)
- 20 packets/second from other session

expected *loss rate* (% packets lost)? $\frac{100 + 20 - 50}{100 + 20} = 58\%$

expected % test flow packets lost? 58%

expected % other session packets lost? 58%

the overloaded switch

let's say switch can handle 50 packets/second

but has:

- 100 packets/second from test flow (checking window size)
- 20 packets/second from other session

expected *loss rate* (% packets lost)? $\frac{100 + 20 - 50}{100 + 20} = 58\%$

expected % test flow packets lost? 58%

expected % other session packets lost? 58%

...but I missed something

a virtuous cycle

what is other session going to when 58% of its packets are lost?

- probably resend them

what about when resent packets are lost?

- probably resent again

if other session doesn't slow down, then...

$$10 \text{ pkt/s} \rightarrow 10 + 58\% \cdot 10 + 58\%^2 \cdot 10 \ldots \approx 48 \text{ pkt/s}$$

the overloaded switch (revised)

let's say switch can handle 50 packets/second

but has:

- 100 packets/second from test flow (checking window size)
- 20 packets/second from other session $\rightarrow$ 48 with resends

expected *loss rate* (% packets lost)? $\frac{100 + 48 - 50}{100 + 48} = 66\%$

expected % test flow packets lost? 66%

expected % other session packets lost? 66%

the overloaded switch (revised)

let's say switch can handle 50 packets/second

but has:

- 100 packets/second from test flow (checking window size)
- 20 packets/second from other session $\rightarrow 48$ with resends

expected *loss rate* (% packets lost)? $\frac{100 + 48 - 50}{100 + 48} = 66\%$

expected % test flow packets lost? 66%

expected % other session packets lost? 66%

- means that 48 pkt/sec is slight underestimate
- though realistically other session should slow down

aside: latency (1)

58% packet loss $\rightarrow$ average packet sent 2.4 times

need one round-trip time (RTT) to detect loss

- probably from duplicate ACK
- if detecting via timeout, probably longer

so need 1.4 RTTs (detecting loss 1.4 times) extra time

mean latency $= \frac{1.4\text{RTTs}}{0.5\text{RTTs}}$ times normal $= 2.8$ times normal

aside: high-percentile latency

58% packet loss

about 10% of time need more than 4 retransmissions

about 5% of the time need more than 5 retransmissions

about 1% of the time need more than 8 retransmissions

sliding windows and retransmissions

assuming that other session doesn't slow down

sliding window approach slows down on losses

sliding window throughput collapse

let's say doing sliding window with 100 packet window

if 1% of the time, we need to resend a packet 8 times, then

probably need around 8 RTTs to send all 100 packets in window

sliding window throughput collapse

let's say doing sliding window with 100 packet window

if 1% of the time, we need to resend a packet 8 times, then

probably need around 8 RTTs to send all 100 packets in window

$\approx$ 8 times slower with same window size

performance v load

slow increase

want to increase *slowly* to avoid overload

original TCP: $+1$ packet/round trip time

slow increase

want to increase *slowly* to avoid overload

original TCP: $+1$ packet/round trip time

$+1$ certainly not optimal choice, but okay heuristic

important theoretically: approx. additive increase

helps ensure good behavior with multiple connections
(we'll talk later about why)

exercise: convergence time (1)

suppose: 50 ms round trip time

initially sending at 600 packets/second
≈ 0.9Mbyte/sec with 1500 byte packets

optimal rate is 10000 packets/second
≈ 15Mbyte/sec with 1500 byte packets

'standard' TCP increase of 1 packet/RTT

how long to get there?

exercise: convergence time (1)

suppose: 50 ms round trip time

initially sending at 600 packets/second
≈ 0.9Mbyte/sec with 1500 byte packets

optimal rate is 10000 packets/second
≈ 15Mbyte/sec with 1500 byte packets

'standard' TCP increase of 1 packet/RTT

how long to get there?

current: 30 packets/RTT ($=$ window size 30)

need to get to: 500 packets/RTT

will take $500 - 30 = 470$ round trips $\approx$ 23500 ms $\approx$ 24 s

fixing bad convergence time

TCP's additive increase is very slow for "high bandwidth-delay" networks

two things make this better:

not in additive increase mode at start of connection

- "slow start" we'll talk about later

more adaptive increase for modern TCP variants

- e.g. FAST TCP, CUBIC TCP, ...
- heuristics to increase faster when appropriate

CUBIC: default congestion control today

default in Linux (since 2006), OS X (since 2014), Windows (since 2019)

- sysadmin has other options they can configure
- can be changed on connection-by-connection basis

big idea: faster increase when further away from window size of last loss

- cubic function with saddle at that window size

intuition:

- search faster if away from "steady state"
- avoid excess losses from 'probing' if at "steady state"

fast decrease

want to decrease quickly to get out of overload

original TCP heuristic: divide by two (minimum 1 packet)

fast decrease

want to decrease quickly to get out of overload

original TCP heuristic: divide by two (minimum 1 packet)

exactly by two probably not important

important theoretically: approx. multiplicative decrease

- will help show okay behavior with multiple flows

AIMD

additive increase + multiplicative decrease

basic of steady-state behavior

non-congestion losses

we were ignoring non-congestion losses

suppose 1% loss rate from transmission errors

if huge bandwidth, 50 ms RTT

with TCP heuristics (+1 packet/RTT, half on loss)...

normal window size? achieved bandwidth (pkts/sec)?

non-congestion losses

we were ignoring non-congestion losses

suppose 1% loss rate from transmission errors

if huge bandwidth, 50 ms RTT

with TCP heuristics (+1 packet/RTT, half on loss)...

normal window size? achieved bandwidth (pkts/sec)?

window size increases for about 100 packets, then halves

starting at window size 8:

8, 9 (17 total), 10 (27), 11 (38), 12 (50), 13 (63), 14 (77), 15 (92), 16 (>100)

→ window size fluctuates from around 8 to 16

non-congestion losses

we were ignoring non-congestion losses

suppose 1% loss rate from transmission errors

if huge bandwidth, 50 ms RTT

with TCP heuristics (+1 packet/RTT, half on loss)...

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starting at window size 8:

8, 9 (17 total), 10 (27), 11 (38), 12 (50), 13 (63), 14 (77), 15 (92), 16 (>100)

→ window size fluctuates from around 8 to 16

12 pkts/50 ms = 240 pkts/sec

non-congestion losses and congestion control

significant non-congestion losses $\rightarrow$ very bad performance with most congestion control

reason why wireless, etc. often does its own acknowledgements and resending

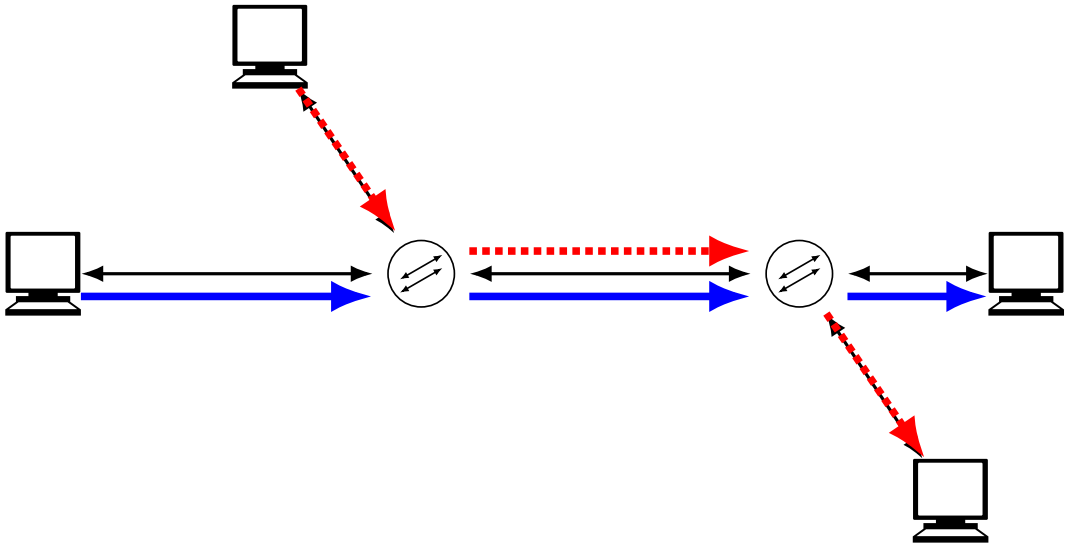
congestion: sharing

want to consider multiple flows

key questions:

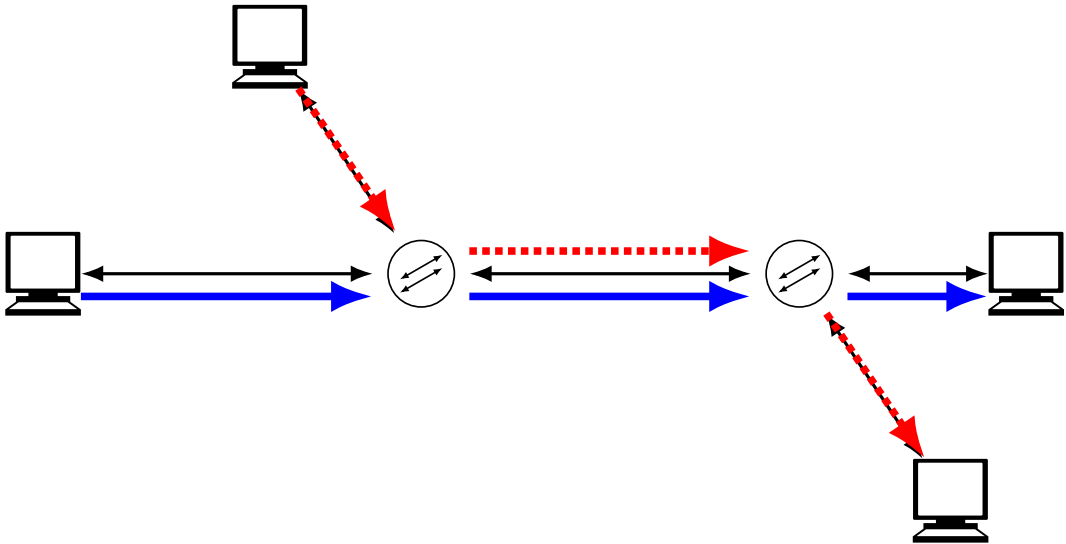
- is it stable if both flows changing window sizes?
- is there one winner/loser?
- is the winner/loser who we want it to be?

exericse: what should happen?



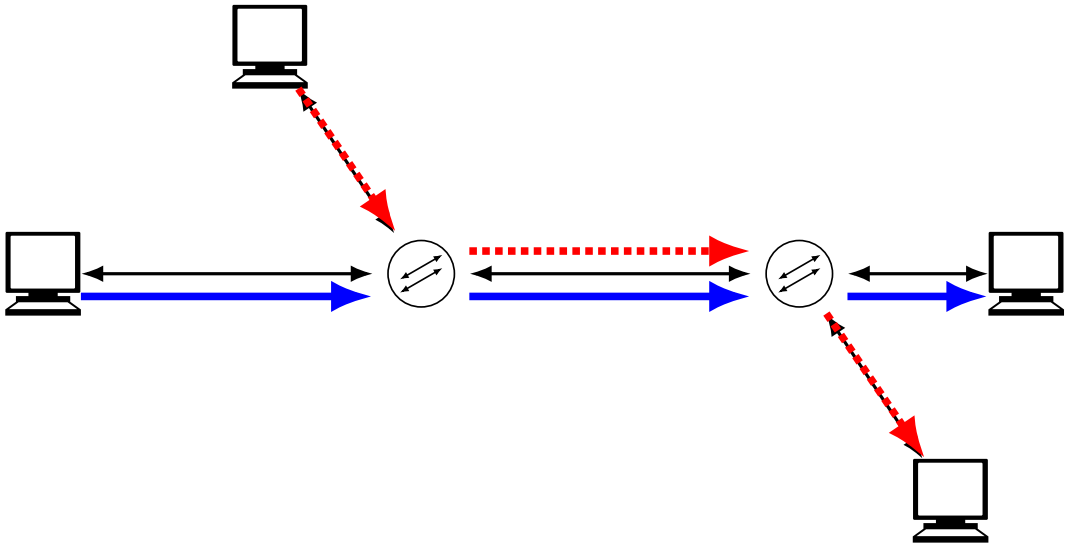
two connections on shared link

exericse: what should happen?



two connections on shared link

exericse: what should happen?



two connections on shared link

showing AIMD

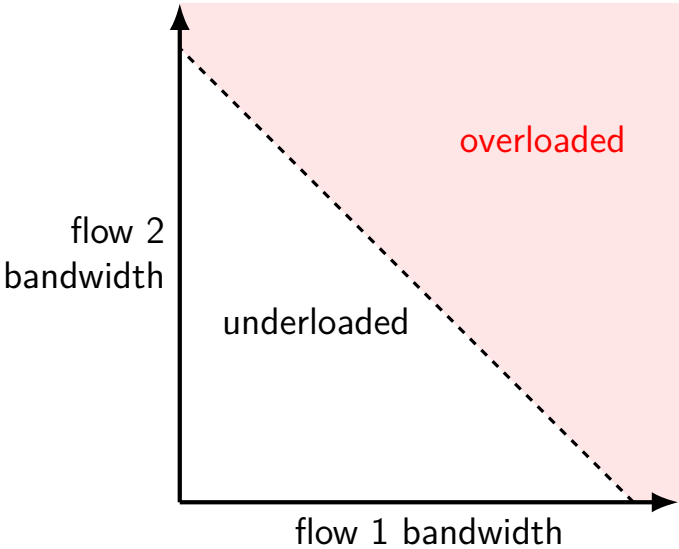
slides based on Chiu and Jain, "Analysis of the Increase and Decrease Algorithms for Congestion Avoidance in Computer Networks"'

1989 paper

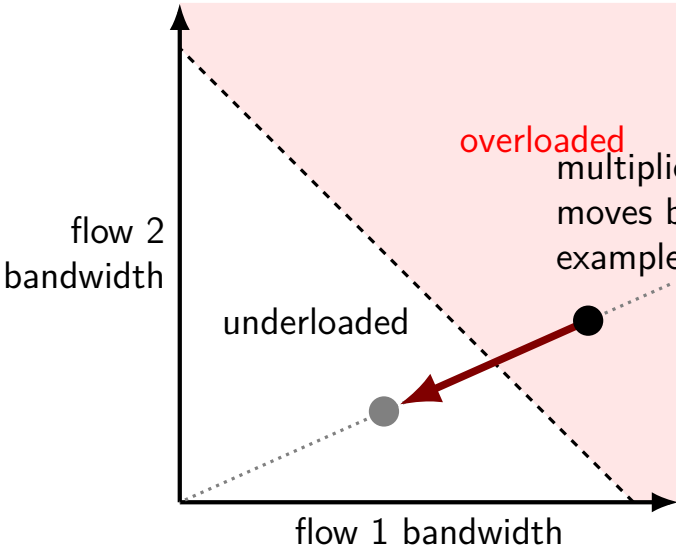
you might notice 1989 is well after TCP was in use

(kinda deployed without all the theory being developed...
...and it's still not really a solved problem)

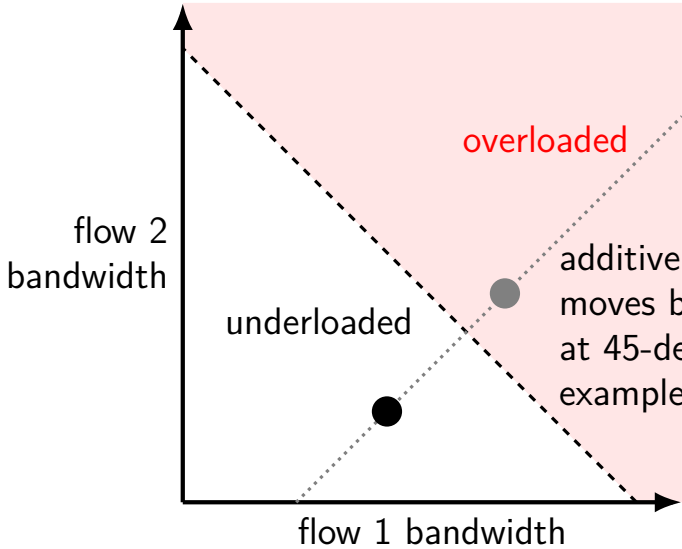
picturing sharing



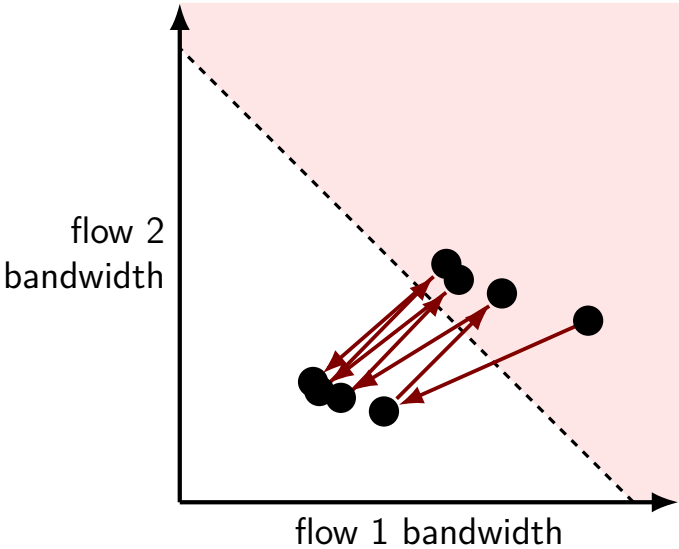
picturing sharing



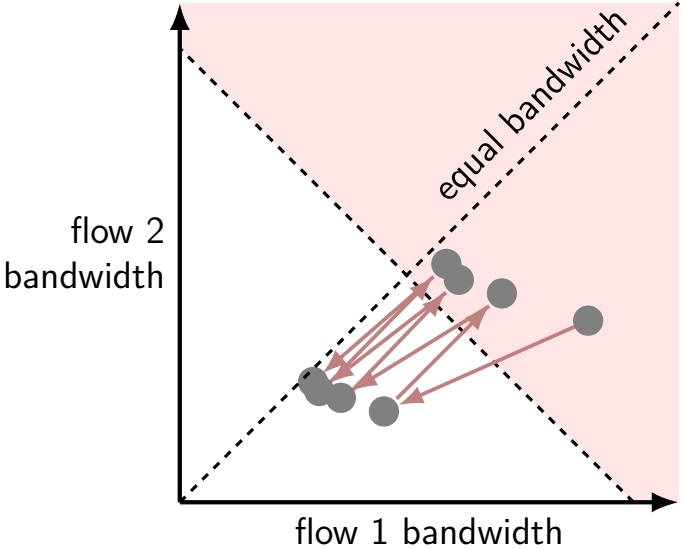
picturing sharing



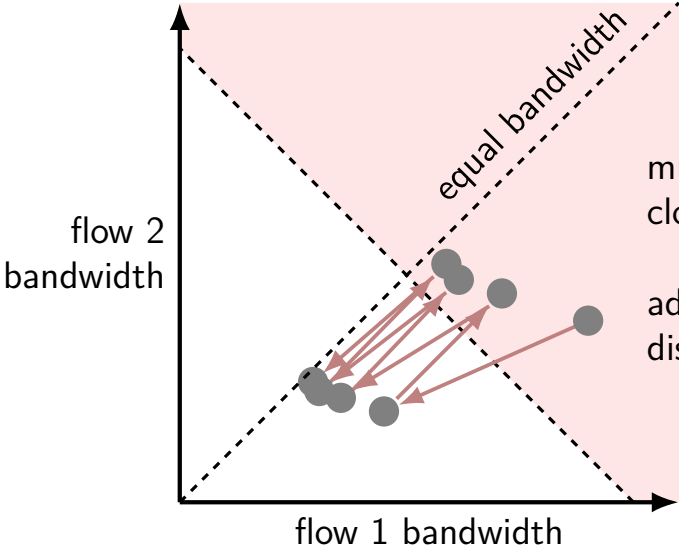
picturing sharing



picturing sharing



picturing sharing



multiplicative decrease brings closer to equal bandwidth line

additive increase keeps same distance from line

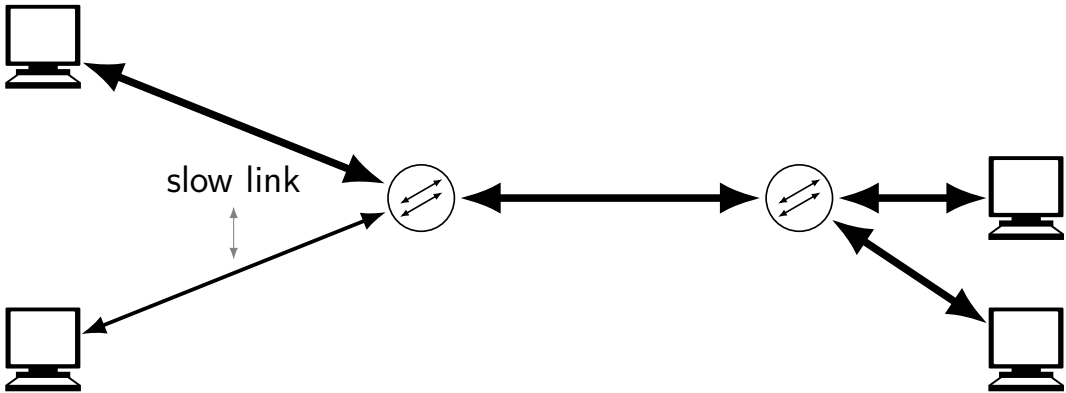
some assumptions we made

...that may not always be true

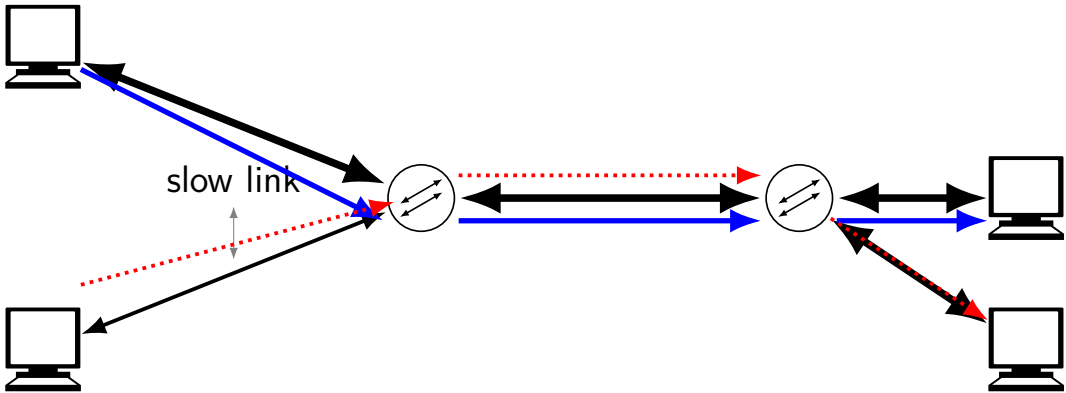
both flows experience drops when network overloaded

same additive increase factor (for 45 degree angle)

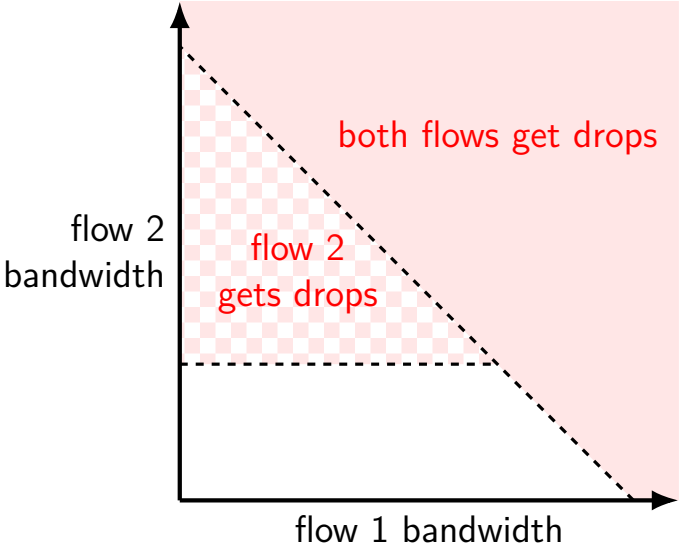
a scenario



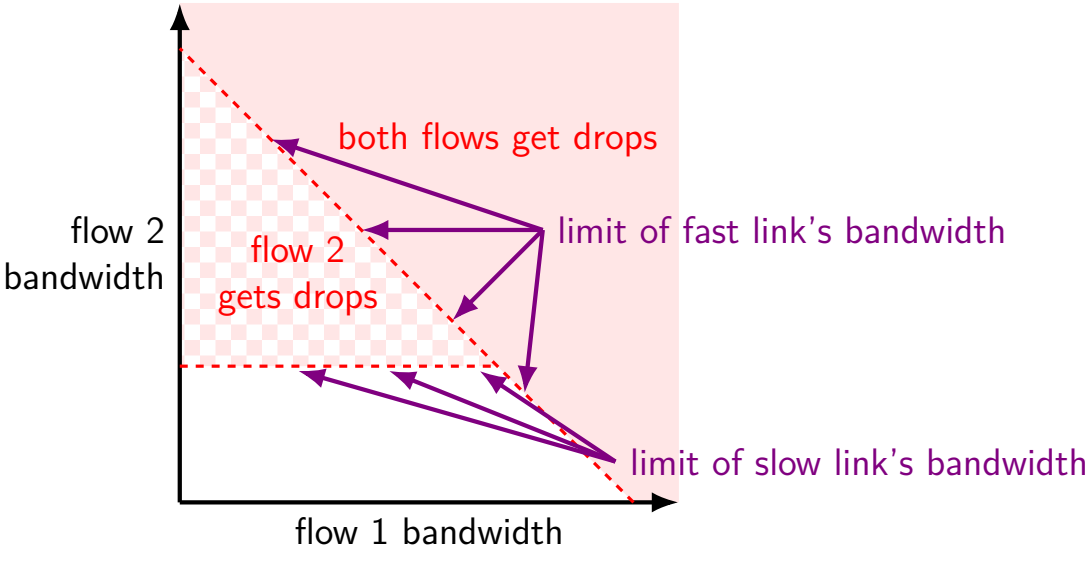
a scenario



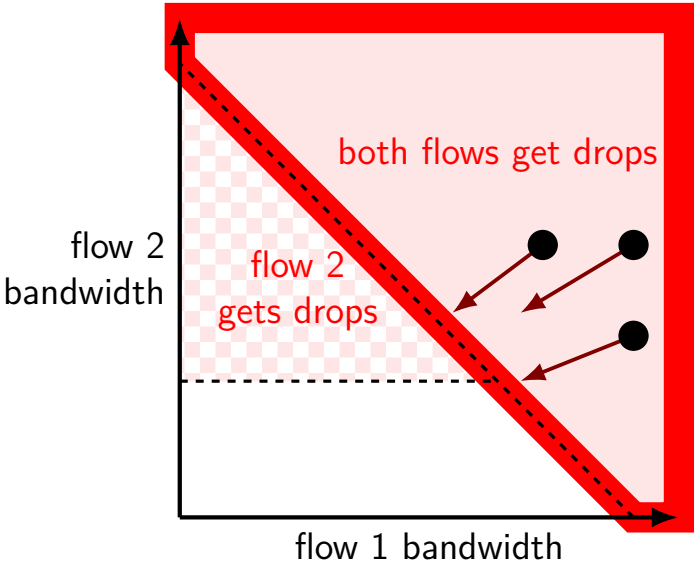
slow link + mixed traffic



slow link + mixed traffic

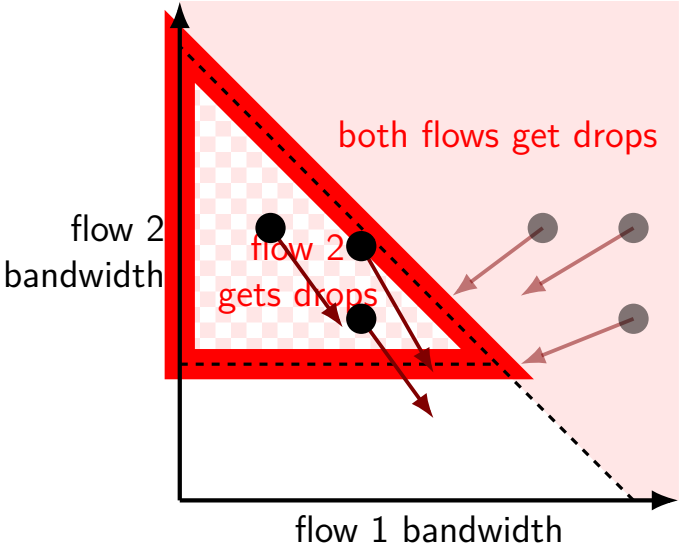


slow link + mixed traffic



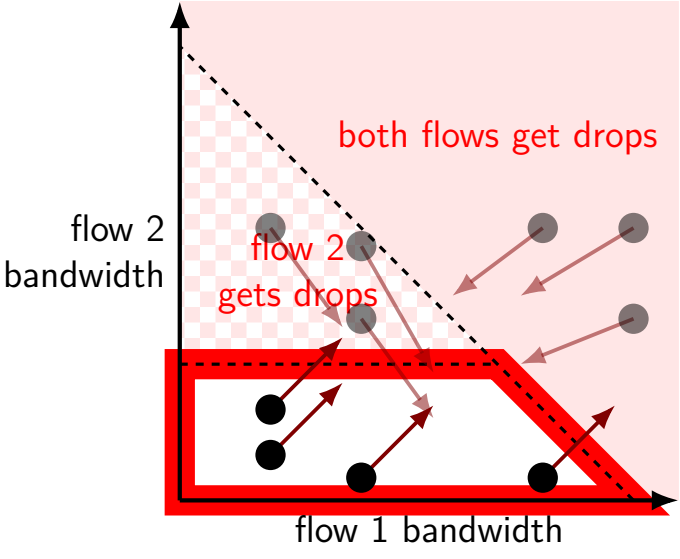
if both flows see drops, multiplicative decrease (toward origin)

slow link + mixed traffic



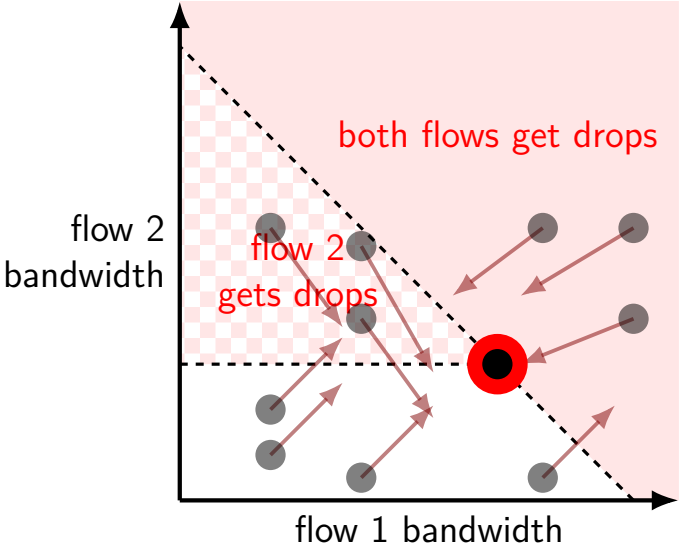
if only flow 2 see drops,
it decreases bandwidth and
flow 1 increase bandwidth

slow link + mixed traffic



if both flows see no drops,
both increase additively
(45 degree angle)

slow link + mixed traffic



result: flow 2 reaches limit of slow link
flow 1 gets the rest of the bandwidth

fairness metrics

would like to say both allocations are ‘fair’

not equal, but most equal we can give on network

would like some way of formalizing this

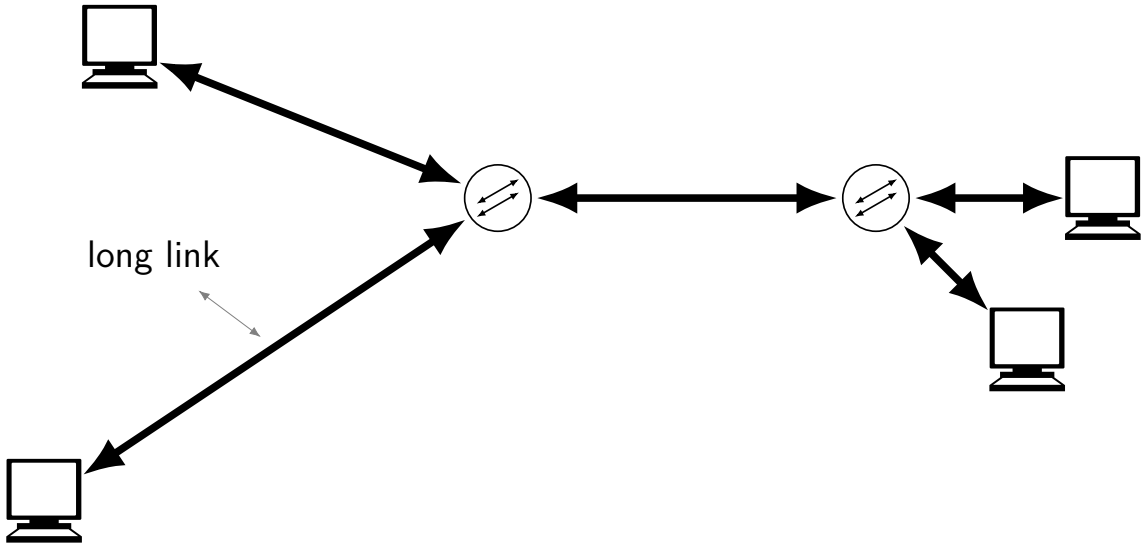
fairness intuition

unfair allocation = someone gets much less than others

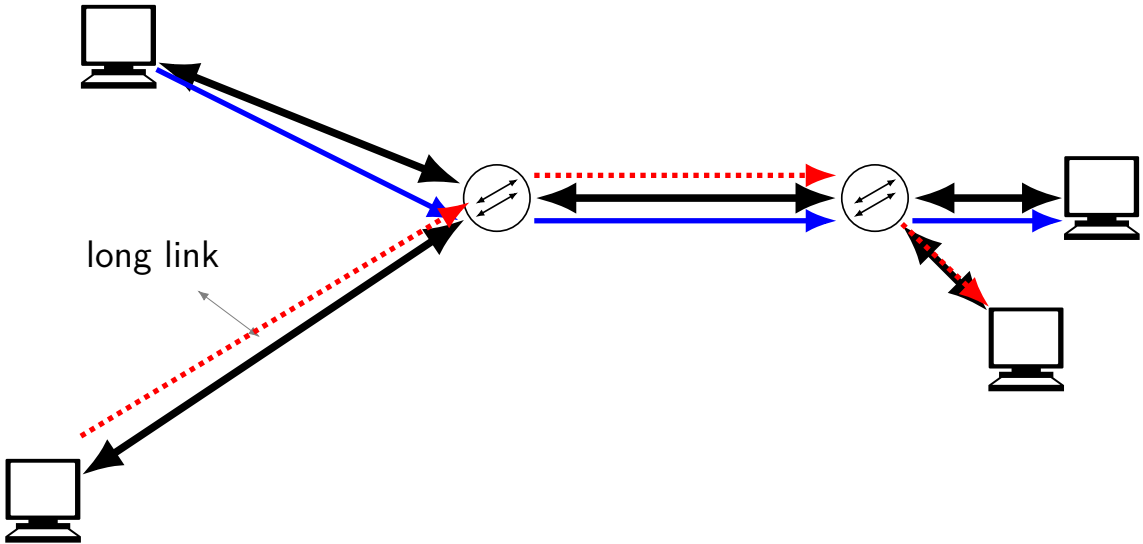
consequence: let's look for "starved" flow

if we can add to starved flow...
without hurting more starved flows...
then that's "unfair"

long links



long links



exercise: window size?

flow 1: 500 packets/sec, 50 ms round trip

flow 2: 500 packets/sec, 100 ms round trip

exercise: what window size achieves this for each flow?

exercise: window size?

flow 1: 500 packets/sec, 50 ms round trip

flow 2: 500 packets/sec, 100 ms round trip

exercise: what window size achieves this for each flow?

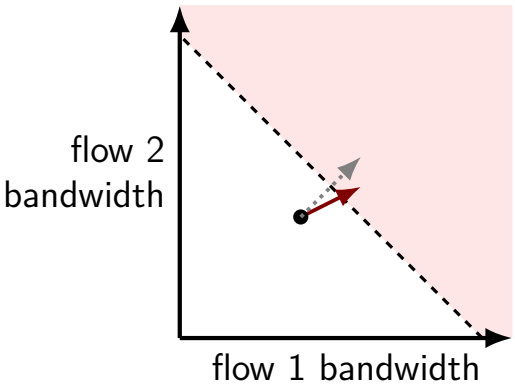
1: window size 25; 2: window size 50

revisiting additive increase

TCP: add +1 to window size each round trip time

flow 1: window 25+1 $\rightarrow$ 26 pkt/50 ms = 520 pkt/sec

flow 2: window 50+1 $\rightarrow$ 51 pkt/100 ms = 510 pkt/sec



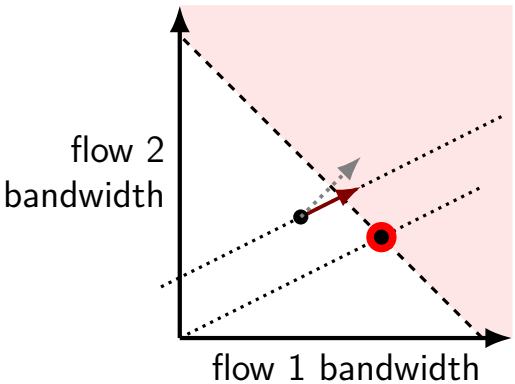
flow 1 increases faster
than flow 2 increases
not 45-degree angle anymore

revisiting additive increase

TCP: add +1 to window size each round trip time

flow 1: window 25+1 → 26 pkt/50 ms = 520 pkt/sec

flow 2: window 50+1 → 51 pkt/100 ms = 510 pkt/sec



in equilibrium
flow 1 gets more bandwidth

other unfairness

lower round-trip gets more bandwidth

can also get more bandwidth by...

using more connections ('independent' windows)

adding more than + 1 packet to window size/RTT

alternate congestion control

lots of changes to congestion control

- some used in modern TCP implementations

on Internet, need to be compatible with "normal" TCP

"TCP-friendly"

- should not make TCP used alongside them behave poorly

fixes from Jacobson's 1987 paper

(*i*) round-trip-time variance estimation

(*ii*) exponential retransmit timer backoff

(*iii*) slow-start

(*iv*) more aggressive receiver ack policy

(*v*) dynamic window sizing on congestion

(*vi*) Karn's clamped retransmit backoff

(*vii*) fast retransmit

fast retransmit

if large window + data packet 2 is lost, then sender will see

ACK 0, ACK 1, ACK 1, ACK 1, ACK 1, ACK 1

duplicate ACKs indicate missing packet 2

shouldn't wait for timeout

fast retransmit

if large window + data packet 2 is lost, then sender will see

ACK 0, ACK 1, ACK 1, ACK 1, ACK 1, ACK 1

duplicate ACKs indicate missing packet 2

shouldn't wait for timeout

→ TCP heuristic: retransmit immediately after ∼3 duplicate ACKs

not 1 duplicate ACK to tolerate some reordering

also some other details (we'll talk later)

fast retransmission

TCP calls this idea of retransmission from duplicate ACKs "fast retransmission"

- was actually not done in early versions of TCP

but problem: what to do with congestion window

solution called 'fast recovery'

self-clocking and dup-ACKs

without losses, sender sends one new packet per ACK

keeps number of packets in network constant

but duplicate ACKs are exception (say window size 6):

recv'd	sent	count of packets in flight
—	data 0-5	6
ACK 0		5
	data 6	6
ACK 1		5
	data 7	6
ACK 1		5
ACK 1		4
ACK 1		3
	data 2	4
ACK 1		3

alternate explanation

sender stopped sending while receiving duplicate ACKs

but we know *most messages got there*

means our usage of network doesn't reflect out window size

TCP's fast retransmission

on third duplicate ACK:

resend packet,

do multiplicative decrease, AND THEN

temporarily add packet to window for each dup ACK

> send packets to replace received packet
> (if allowed by multiplicative-decreased window)

reset window size back when 'new' ACK

self-clocking and fast retransmit

adjust window size to keep packets in flight constant:

recv'd	sent	count packets in flight	send window size (range)
—	data 0-5	6	6 (0-5)
ACK 0		5	6 (1-6)
	data 6	6	6 (1-6)
ACK 1		5	6 (2-7)
	data 7	6	6 (2-7)
ACK 1		5	6 (2-7)
ACK 1		4	6 (2-7)
ACK 1		3	6 (2-7)
	data 2	4	8 (2-9)
	data 8	5	8 (2-9)
	data 9	6	8 (2-9)
ACK 1		5	9 (2-10)
1	data 10	5	9 (2-10)

fixes from Jacobson's 1987 paper

(*i*) round-trip-time variance estimation

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“slow start”

not very well named

problem is that additive increase doesn't find capacity quickly

exponential(ish) increase to find *initial* window size

“slow start” on connection begin

set window size = 1 packet

increase by one packet for each ACK

...until first packet loss

then revert to additive increase

actually slower at increasing

“slow start” later

keep track of window size after multiplicative decrease

called `ssthresh`

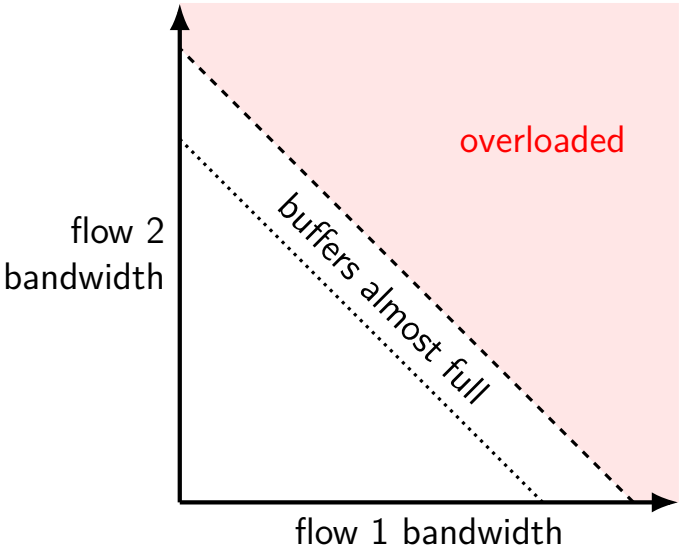
- probably variable name in BSD code for this

on timeout (not dupack), reset window size to 1 packet, BUT

use ‘slow start’ until ssthresh reached or more congestion

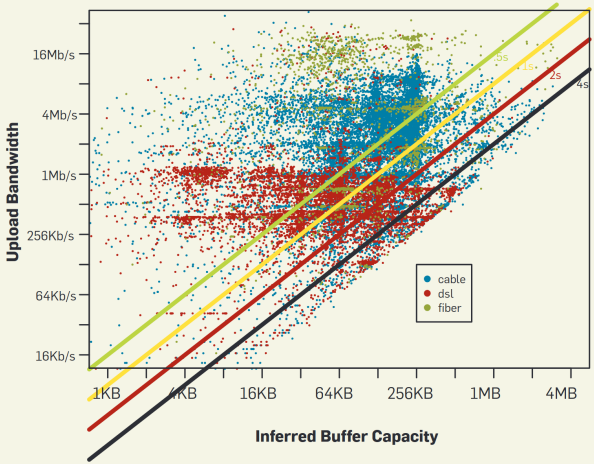
- backoff network more aggressively, and
- hopefully find correct new window size faster?

filling buffers



big buffers? (in 2011 or so)

Figure 5. Plot reproduced from ICSI's Netalyzr studies.



Jim Gettys and Kathleen Nichols,

"Bufferbloat: Dark Buffers in the Internet" (CACM, Jan 2012)

problems with big buffers

high latency — bad for some applications

slower response to congestion

- 1 second round trip time = 1 second to detect congestion
- more likely to have ‘congestion collapse’

avoiding big buffers

multiple fixes (that can be combined):

use smaller buffers?

- simpliest solution

detect congestion without full buffer...

- by choosing when/which packets to drop better?
- by using something other than drops?

avoiding big buffers

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also some other advantages to new ways of detecting congestion

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detect congestion without full buffer...

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also some other advantages to new ways of detecting congestion

other congestion signals

so far: detecting congestion via drops

- need data to go missing
- transmitting redundant data
- filling up buffers causing high latency

some alternate ideas:

have switches/routers ‘mark’ packets

latency from longer queues

ECN

explicit congestion notification

when buffer 'close' to full

switches/router set 'congestion experienced' (CE) bit in some packets

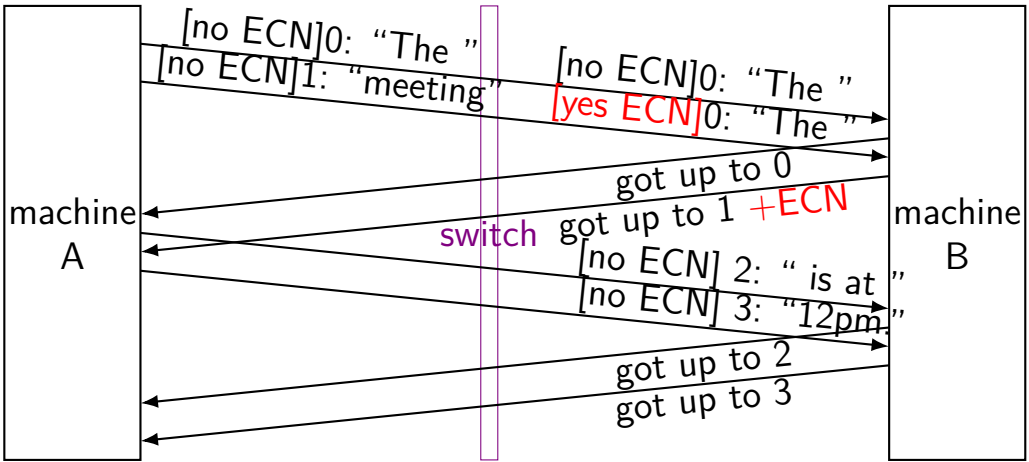
- field in IP header

CE bit echoed back in ACKs

for compatibility:

- sender opts-in to ECN by setting field in IP packet
- if not set, switch/router just drops packet

ECN timeline



data sent has place for ECN bit to be placed

switch modifies ECN bit if buffer close to full

reacting to ECN marks

multiple options for using ECN marks

simplest idea:

adjust window as if packet was dropped

...but don't need to resend data

example: DCTCP

DataCenter TCP (2010)

intended for datacenters

- high bandwidth, low latency networks

based on explicit congestion notification

...but uses different multiplicative decrease strategy

measure portion of packets marked recently α

decrease by factor of $1 - \alpha/2$

- respond gradually to congestion
- start responding early (packets marked when queues far from full)

very different congestion control

fuller queues $\rightarrow$ higher latency

fuller queues $\rightarrow$ throughput same as window increases

very different congestion control

fuller queues $\rightarrow$ higher latency

fuller queues $\rightarrow$ throughput same as window increases

strategy: monitor throughput/latency to detect full queues

goal: fill link without making queue grow (much) in size

‘Vegas’-style congestion control

record “base” round-trip time

- connection start or lowest observed

“ideal” throughput should be one window / base round trip time

- (Vegas paper calls this “expected” throughput)
- what would happen with no queuing delay

goal: control what “ideal” - actual throughput is

- if 0, queues are probably empty, can increase window
- if large, queues are too big, decrease window

BBR-style congestion control

if queues are empty, larger window:

latency stays the same and throughput increases

if queues are filling, larger window:

throughput stays the same and latency increases

BBR-style congestion control

if queues are empty, larger window:

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BBR-style congestion control

if queues are empty, larger window:

latency stays the same and throughput increases

if queues are filling, larger window:

throughput stays the same and latency increases

observe effect of sending more/fewer packets periodically

estimate 'boundary' based on observed latency/throughput

keep window size near boundary most of the time

backup slides